

The constant $A(p)$

Quasi-stationary amplitude, closed forms, and the Koenigs obstruction

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Chapter 2

The constant $A(p)$

2.1 Introduction

For the binary Galton–Watson process with division probability $p \in [0, \frac{1}{2})$ the survival probability decays geometrically, $S_n \sim A(p)(2p)^n$, and the mean population conditional on non-extinction converges to $1/A(p)$. That much was established in the mathematical introduction. Everything the limiting conditional law does — its mean, its variance, its dependence on how close the process sits to criticality — passes through the single constant

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}, \quad p \in [0, \tfrac{1}{2}). \quad (2.1)$$

This chapter is about that constant: what it is, how to compute it, why no elementary formula for it has been found despite a determined search, and exactly what can and cannot be proved about the failure of that search. The results are original to this thesis except where attributed.

Approximate values are easy enough to obtain. Iterate $S_{n+1} = 2pS_n - pS_n^2$ until S_n is very nearly stationary, then form $S_n/(2p)^n$. Repeating over a range of p traces the curve of [figure 2.2](#), and this works well over most of the interval. It fails as $p \rightarrow \frac{1}{2}^-$, because the number of generations needed for S_n to reach near-stationarity grows without bound — from around 10^2 at $p = 0.2$ to nearly 5×10^5 at $p = 0.4999$.

The argument descends from the concrete to the structural, and each stage constrains the one after it. After a short recapitulation of the setup inherited from the mathematical introduction ([section 2.2](#)), two exact representations come first. Rewriting the survival recursion as a telescoping product ([section 2.3](#)) proves that the limit (2.1) exists and is strictly positive; converting that product into a series ([section 2.4](#)) isolates the geometric decay in a factor carrying no dynamics at all, so that two-sided bounds can be read off by inspection. Those bounds trap $A(p)$ between half the continuous-time constant $A_c(p)$ and a companion upper bound, and a parity argument caps it at $\frac{1}{2}$. The same series then yields the near-critical behaviour $A(p) = 2\varepsilon[1 + O(\varepsilon \log(1/\varepsilon))]$, with $\varepsilon = 1 - 2p$ ([section 2.5](#)), locating the logarithm that makes convergence to the leading term slow, and with it the exact relationship between A and A_c ([section 2.6](#)).

What none of that yields is a formula. [Section 2.7](#) records the search for one and the manner of its failure, which turns out to be diagnostic rather than merely discouraging: what gave way was not the ingenuity of the search but the assumption that there was a formula to be found. The explanation is dynamical. Under the substitution $S_n = 2w_n$ the survival recursion becomes the logistic map, and $A(p)$ is twice a value of that map’s Koenigs linearising function ([section 2.8](#)); for every multiplier in the subcritical window that function is hypertranscendental ([section 2.9](#)), so no

elementary expression in its own variable can be evaluated to produce $A(p)$. [Section 2.10](#) delimits what this does and does not settle and reports the inverse-symbolic evidence bearing on what it leaves open, and [sections 2.12](#) and [2.13](#) draw the practical and the residual conclusions.

A word on why the scope section exists. There are at least four inequivalent things one might mean by “ $A(p)$ has no closed form”, and a theorem about one of them is not a theorem about another. What is proved here concerns the conjugacy as a function of its own dynamical variable. It does not decide the arithmetic nature of the individual numbers $A(p_0)$, nor whether the map $p \mapsto A(p)$ is elementary. Both remain open, and the chapter says so wherever the distinction could otherwise be blurred.

2.2 Standing setup

This section fixes what is carried over from the mathematical introduction. It is deliberately short: the probabilistic development belongs there, and only the objects this chapter operates on are restated.

Let Z_n be the binary Galton–Watson process started from a single progenitor, $Z_0 = 1$, in which each individual independently divides into two with probability p and dies with probability $1 - p$ at each generational increment. Its offspring generating function is $\phi(z) = pz^2 + (1 - p)$, the mean population is $\mathbb{E}(Z_n) = (2p)^n$, and the survival probability $S_n = \mathbb{P}(Z_n > 0)$ satisfies

$$S_{n+1} = 2pS_n - pS_n^2, \quad S_0 = 1. \quad (2.2)$$

Throughout this chapter p is confined to the subcritical range

$$0 \leq p < \frac{1}{2}, \quad (2.3)$$

where extinction is certain and $S_n \downarrow 0$. Two derived symbols are used constantly:

$$\varepsilon = 1 - 2p \in (0, 1] \quad (\text{the distance from criticality}), \quad r = 2p \in [0, 1) \quad (\text{the logistic multiplier}). \quad (2.4)$$

Because the quadratic term in (2.2) becomes negligible once S_n is small, the decay is asymptotically geometric with ratio $2p$, and the constant $A(p)$ of (2.1) is the amplitude of that decay:

$$S_n \sim A(p) (2p)^n \quad (n \rightarrow \infty). \quad (2.5)$$

The relation is an asymptotic equivalence, not an identity: S_n itself tends to zero, and what converges is the ratio $S_n/(2p)^n$. That this ratio converges to something finite and strictly positive is not assumed here; it is proved in [section 2.3](#). The constant is not accessible to any argument confined to late generations, because it is precisely the accumulated effect of the early nonlinear corrections, before the recursion attenuates to $S_{n+1} \approx 2pS_n$.

Two consequences of (2.5) are the reason the constant matters. Splitting $\mathbb{E}(Z_n)$ over the events $\{Z_n = 0\}$ and $\{Z_n > 0\}$, the first contributing nothing, gives $\mathbb{E}(Z_n | Z_n > 0) = (2p)^n/S_n$ and hence

$$\mathbb{E}(Z_n | Z_n > 0) \longrightarrow \frac{1}{A(p)}, \quad (2.6)$$

the *quasi-stationary mean*; and the corresponding second-moment calculation gives a limiting conditional variance

$$\text{Var}(Z_n | Z_n > 0) \longrightarrow \frac{1}{A(p)} \left(1 + \frac{1}{1 - 2p} \right) - \frac{1}{A(p)^2}, \quad (2.7)$$

again a function of p and $A(p)$ alone. The derivations of (2.6) and (2.7), together with the existence of the Yaglom limit they describe, are given in the mathematical introduction and are not repeated.

One comparison object is also inherited. The continuous-time linear birth–death process with per-capita birth rate λ and death rate $\mu > \lambda$ has a limiting conditional mean $\mu/(\mu - \lambda)$, and matching the parametrisations through the competing-clocks identity $p = \lambda/(\lambda + \mu)$ — proved in the mathematical introduction and used here by citation — turns this into

$$A_c(p) = \frac{1 - 2p}{1 - p} = \frac{2\varepsilon}{1 + \varepsilon}, \quad \frac{1}{A_c(p)} = \frac{1 - p}{1 - 2p}. \quad (2.8)$$

Unlike $A(p)$, the continuous-time constant $A_c(p)$ is elementary. Much of what follows is an account of the exact relationship between the two, and of why the discrete one is harder.

Finally, an endpoint convention. At $p = 0$ the process dies at the first step and (2.2) degenerates; we set

$$A(0) := \lim_{p \downarrow 0} A(p) = \frac{1}{2}, \quad (2.9)$$

a value confirmed independently by the product of [section 2.3](#) and by the parity bound of [proposition 2.4](#).

2.3 An infinite-product representation

Take the nonlinear map for the survival probability,

$$S_{n+1} = 2pS_n - pS_n^2, \quad S_0 = 1,$$

and substitute $S_n = 2w_n$. This gives

$$w_{n+1} = 2p w_n(1 - w_n) = r w_n(1 - w_n), \quad r = 2p, \quad w_0 = \frac{1}{2}, \quad (2.10)$$

which is the logistic map — the same map that exhibits the period-doubling route to chaos as r increases from 0 to 4 ([figure 2.1](#)). The subcritical window is $r \in [0, 1)$, at the extreme left of that picture, where the origin is a globally attracting fixed point. That this innocuous corner of the logistic family is nevertheless enough to obstruct a closed form is the substance of [sections 2.8](#) and [2.9](#).

In terms of w , (2.1) reads

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n} = 2 \lim_{n \rightarrow \infty} \frac{w_n}{r^n}. \quad (2.11)$$

Any sequence can be written as a telescoping product,

$$x_n = x_0 \left(\frac{x_1}{x_0} \right) \left(\frac{x_2}{x_1} \right) \cdots \left(\frac{x_n}{x_{n-1}} \right) = x_0 \prod_{k=0}^{n-1} \frac{x_{k+1}}{x_k},$$

and applying this with $x_n = w_n/r^n$ the ratio simplifies at once:

$$\frac{x_{k+1}}{x_k} = \frac{w_{k+1}}{w_k} \cdot \frac{1}{r} = \frac{r w_k(1 - w_k)}{w_k} \cdot \frac{1}{r} = 1 - w_k. \quad (2.12)$$

The whole of the nonlinearity has been isolated into a single factor per generation, and the geometric decay has cancelled out entirely.

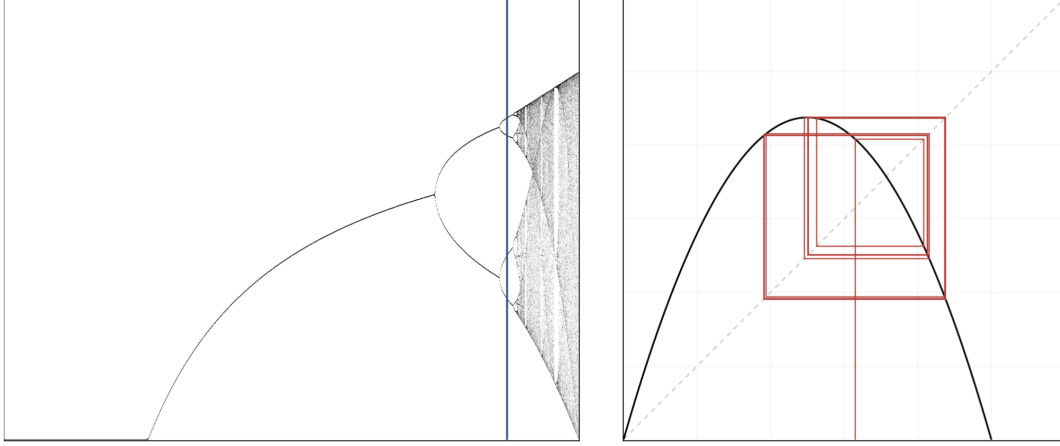


Figure 2.1: Bifurcation diagram of the logistic map $w \mapsto rw(1 - w)$, showing the period doubling of stable states and the onset of chaos. The blue sample line on the left is at $r = 3.5255$; the right-hand plot shows a trajectory of the map at the same r . The regime relevant to subcritical branching is $r = 2p \in [0, 1)$, off the left-hand edge of this diagram, where the dynamics are as simple as they can be.

Proposition 2.1 (Product representation). *For $p \in (0, \frac{1}{2})$, with S_n generated by (2.2) and $w_k = S_k/2$,*

$$A(p) = \prod_{k=0}^{\infty} (1 - w_k) = \frac{1}{2} \prod_{k=1}^{\infty} \left(1 - \frac{S_k}{2}\right), \quad (2.13)$$

and the product converges to a strictly positive limit. In particular the limit (2.1) defining $A(p)$ exists, and $0 < A(p) < \frac{1}{2}$.

Proof. Iterating (2.12) gives $x_n = x_0 \prod_{k=0}^{n-1} (1 - w_k)$ with $x_0 = w_0 = \frac{1}{2}$. For $0 < w_k < 1$ the infinite product $\prod_{k \geq 1} (1 - w_k)$ converges to a strictly positive limit if and only if $\sum_{k \geq 1} w_k < \infty$. From (2.10) with $0 < r < 1$ and $0 < w_n < 1$ we have $w_{n+1} < rw_n$, so by induction $w_n \leq w_0 r^n$; since $0 < r < 1$ this bound is summable, and the product converges to a positive finite value. Hence $\lim_n x_n$ exists and is positive, and by (2.11),

$$A(p) = 2 \lim_n x_n = 2 \cdot \frac{1}{2} \prod_{k=0}^{\infty} (1 - w_k) = \prod_{k=0}^{\infty} (1 - w_k).$$

Pulling out the factor $1 - w_0 = \frac{1}{2}$ gives the second form, and since every remaining factor lies strictly in $(0, 1)$ for $p > 0$, the value is strictly below $\frac{1}{2}$. $\square$

2.3.1 A certified truncation

The partial products

$$A_N(p) = \frac{1}{2} \prod_{k=1}^N \left(1 - \frac{S_k}{2}\right) \quad (2.14)$$

decrease monotonically to $A(p)$, so every truncation is a rigorous upper bound. A matching lower bound follows from the elementary inequality $\prod_k (1 - a_k) \geq 1 - \sum_k a_k$ for $a_k \in [0, 1]$ applied to the

discarded tail. Since $w_{k+1} \leq rw_k$, the tail sum is dominated by a geometric series and

$$A_N(p) \left(1 - \frac{w_{N+1}}{1-r}\right) \leq A(p) \leq A_N(p), \quad (2.15)$$

whenever the left-hand factor is positive. This is worth having: it converts the product into a computation with a certificate, and gives a stopping criterion stronger than the usual observation that successive floating-point iterates have stopped changing.

2.3.2 Is this an advance?

Only a modest one. (2.13) is exact, it proves existence, and by (2.15) it brackets $A(p)$ at every order, which the raw ratio $S_n/(2p)^n$ does not. But computing the factors still requires iterating the nonlinear map to generate the S_k , so the near-critical difficulty is untouched: as $r \rightarrow 1$ the bound $w_n \leq w_0 r^n$ degrades and the number of factors needed grows without limit. What the product does supply is a route to the sharper result of the next section, where the same telescoping idea applied to the reciprocal sequence yields something genuinely more informative.

2.4 An exact series and two-sided bounds

The product converts into a telescoping sum, and the sum is much more informative. Write $r = 2p$ and note that the recursion factorises as

$$S_{n+1} = r S_n \left(1 - \frac{S_n}{2}\right).$$

Define $v_n = r^n/S_n$, so that $v_n \rightarrow 1/A(p)$ by (2.1). Then

$$v_{n+1} - v_n = \frac{r^{n+1}}{r S_n \left(1 - \frac{S_n}{2}\right)} - \frac{r^n}{S_n} = \frac{r^n}{S_n} \left[\frac{1}{1 - \frac{S_n}{2}} - 1 \right] = \frac{r^n}{S_n} \cdot \frac{S_n/2}{1 - \frac{S_n}{2}} = \frac{r^n}{2 - S_n}. \quad (2.16)$$

Summing from $v_0 = 1/S_0 = 1$ gives an exact representation of the quasi-stationary mean.

Proposition 2.2 (Series representation). *For $p \in [0, \frac{1}{2})$, with S_n generated by (2.2),*

$$\frac{1}{A(p)} = 1 + \sum_{n=0}^{\infty} \frac{(2p)^n}{2 - S_n}. \quad (2.17)$$

Unlike the product, this representation isolates the geometric decay in a factor that does not depend on the dynamics at all; the whole of the nonlinearity sits in the bounded denominator $2 - S_n$. That is what makes it useful, because S_n is confined to $(0, 1]$ and therefore $2 - S_n$ to $[1, 2)$. Bounding the denominator by its extremes and summing the resulting geometric series gives the following.

Proposition 2.3 (Two-sided bounds). *With $\varepsilon = 1 - 2p$ as in (2.4), for $p \in (0, \frac{1}{2})$,*

$$\frac{\varepsilon}{1 + \varepsilon} < A(p) < \frac{2\varepsilon}{1 + 2\varepsilon}, \quad (2.18)$$

that is,

$$\frac{1 - 2p}{2(1 - p)} < A(p) < \frac{2(1 - 2p)}{3 - 4p}. \quad (2.19)$$

The lower bound is exactly half the continuous-time constant (2.8):

$$A(p) > \frac{1}{2} A_c(p). \quad (2.20)$$

Proof. Since $0 < S_n \leq 1$ for all n we have $1 \leq 2 - S_n < 2$, and hence $\frac{1}{2}r^n < r^n/(2 - S_n) \leq r^n$, with equality on the right only at $n = 0$, where $S_0 = 1$. Summing over $n \geq 0$ and using $\sum_n r^n = 1/(1 - r) = 1/\varepsilon$,

$$1 + \frac{1}{2\varepsilon} < \frac{1}{A(p)} < 1 + \frac{1}{\varepsilon},$$

and inverting gives (2.18). Substituting $\varepsilon = 1 - 2p$ gives (2.19), whose lower bound is $(1 - 2p)/(2(1 - p)) = \frac{1}{2}A_c(p)$ by (2.8). $\square$

Two features of [proposition 2.3](#) deserve comment, because between them they account for the whole shape of the discrete curve.

At the far subcritical end, $p \rightarrow 0$, we have $\varepsilon \rightarrow 1$ and the lower bound tends to $\frac{1}{2}$. Since $A(p) \leq \frac{1}{2}$ by [proposition 2.4](#) below, the bound is asymptotically attained and $A(0^+) = \frac{1}{2}$, confirming the endpoint convention (2.9). At the critical end, $p \rightarrow \frac{1}{2}^-$, we have $\varepsilon \rightarrow 0$, the lower bound behaves like ε and the upper like 2ε ; [section 2.5](#) shows that it is the upper bound that is attained there. So $A(p)$ runs from its lower bound at one end of the interval to its upper bound at the other, which is why no constant rescaling of A_c can reproduce it.

Proposition 2.4 (Parity bound). *For the binary Galton–Watson process started from a single individual, $A(p) \leq \frac{1}{2}$ for all $p \in [0, \frac{1}{2})$, with equality approached as $p \rightarrow 0$.*

Proof. Every individual leaves either 0 or 2 offspring, so Z_n is even for every $n \geq 1$. Conditional on survival, therefore, $Z_n \geq 2$, and hence $\mathbb{E}(Z_n | Z_n > 0) \geq 2$ for all $n \geq 1$. Letting $n \rightarrow \infty$ and using (2.6) gives $1/A(p) \geq 2$. As $p \rightarrow 0$ the conditional population is 2 with probability tending to one, so the bound is attained in the limit. $\square$

The parity bound is a genuinely different kind of constraint from [proposition 2.3](#): it comes from the arithmetic of the offspring law rather than from the analysis of the recursion, and it is what fixes the left-hand endpoint. [Table 2.1](#) collects numerical values against all three bounds.

2.5 Behaviour as $p \rightarrow \frac{1}{2}^-$

The bounds (2.18) show that $A(p)$ is of order ε near criticality but leave a factor of two unresolved. The series (2.17) settles it, and does so more transparently than the matching argument one is first tempted to try. It also identifies the order of the first relative correction, which turns out to carry a logarithm and is the reason the leading asymptotic is a poor surrogate except very close to $p = \frac{1}{2}$.

Theorem 2.5 (Near-critical amplitude). *As $p \uparrow \frac{1}{2}$, with $\varepsilon = 1 - 2p \downarrow 0$,*

$$A(p) = 2\varepsilon \left[1 + O\left(\varepsilon \log \frac{1}{\varepsilon}\right) \right]. \quad (2.21)$$

In particular

$$A(p) \sim 2(1 - 2p), \quad \frac{1}{A(p)} \sim \frac{1}{2(1 - 2p)}. \quad (2.22)$$

Proof. Write $r = 1 - \varepsilon$. The identity

$$\frac{1}{2 - S_n} = \frac{1}{2} + \frac{S_n}{2(2 - S_n)} \quad (2.23)$$

Table 2.1: The constant $A(p)$ computed from the product (2.13) in 60-digit arithmetic, against the bounds of proposition 2.3 and the near-critical asymptotic $2(1-2p)$ of theorem 2.5. The bounds are loose in the middle of the range and tighten at both ends: at $p = 0.05$ the lower bound is already within 3% of A , whereas at $p = 0.4999$ it is the upper bound that is within 0.1%. The last two columns give the quasi-stationary mean and the number of factors of (2.13) needed for the product to reach full precision — the cost that section 2.12 has to manage.

p	$\frac{1}{2}A_c(p)$	$A(p)$	$\frac{2(1-2p)}{3-4p}$	$2(1-2p)$	$1/A(p)$	iterations
0.05	0.473684	0.486180	0.642857	1.800000	2.0568	45
0.1	0.444444	0.469382	0.615385	1.600000	2.1305	64
0.2	0.375000	0.423895	0.545455	1.200000	2.3591	113
0.3	0.285714	0.353967	0.444444	0.800000	2.8251	201
0.4	0.166667	0.237647	0.285714	0.400000	4.2079	458
0.45	0.090909	0.145069	0.166667	0.200000	6.8933	966
0.48	0.038462	0.068011	0.074074	0.080000	14.7036	2 473
0.49	0.019608	0.036395	0.038462	0.040000	27.4764	4 965
0.499	0.001996	0.003944	0.003984	0.004000	253.519	48 992
0.4999	0.000200	0.000399	0.000400	0.000400	2504.65	478 905

turns the exact series (2.17) into

$$\frac{1}{A(p)} = 1 + \frac{1}{2\varepsilon} + D_\varepsilon, \quad D_\varepsilon := \sum_{n=0}^{\infty} r^n \frac{S_n}{2(2-S_n)}, \quad (2.24)$$

the geometric sum $\sum_n r^n = 1/\varepsilon$ having been evaluated. Only a logarithmic bound on the remainder D_ε is needed.

The map $s \mapsto ps(2-s)$ is increasing in p and in s on $s \in [0, 1]$, so comparing (2.2) at parameter p with the same recursion at $p = \frac{1}{2}$ from the common initial value $S_0 = 1$ gives, by induction, $S_n(p) \leq S_n(\frac{1}{2})$ for every n . At criticality the reciprocal increments satisfy

$$\frac{1}{S_{n+1}(\frac{1}{2})} - \frac{1}{S_n(\frac{1}{2})} = \frac{1}{2(1-S_n(\frac{1}{2})/2)} \geq \frac{1}{2}, \quad (2.25)$$

so $1/S_n(\frac{1}{2}) \geq 1 + n/2$ and hence $S_n(\frac{1}{2}) \leq 2/(n+2)$. Since $2 - S_n \geq 1$, these two facts combine to give

$$\begin{aligned} 0 \leq D_\varepsilon &\leq \frac{1}{2} \sum_{n=0}^{\infty} r^n S_n(p) \leq \frac{1}{2} \sum_{n=0}^{\infty} r^n \frac{2}{n+2} = \sum_{n=0}^{\infty} \frac{r^n}{n+2} \\ &= \frac{-\log(1-r) - r}{r^2} = O\left(\log \frac{1}{\varepsilon}\right). \end{aligned} \quad (2.26)$$

Factoring $1/(2\varepsilon)$ out of (2.24),

$$\frac{1}{A(p)} = \frac{1}{2\varepsilon} \left[1 + 2\varepsilon(1 + D_\varepsilon) \right],$$

and the quantity following the leading 1 is $O(\varepsilon \log(1/\varepsilon))$. Inverting gives (2.21). $\square$

The proof locates the logarithm precisely: it comes from the near-critical decay $S_n \lesssim 2/(n+2)$ persisting across the whole range $n \lesssim 1/\varepsilon$ over which the geometric factor r^n still carries weight. The heuristic version of the same computation is worth seeing, because it makes the mechanism visible. The summand $r^n/(2-S_n)$ involves two decays on very different scales: the geometric factor $r^n = (1-\varepsilon)^n$ decays on the scale $n \sim 1/\varepsilon$, which is large, whereas S_n is close to its critical behaviour $2/n$ and has already fallen to near zero after $O(1)$ generations. For the overwhelming majority of the terms that carry weight, therefore, $S_n \approx 0$ and the summand is approximately $r^n/2$; summing gives $1/A \approx 1/(2\varepsilon)$ at once. The rigorous argument above is that estimate with the neglected mass bounded rather than discarded.

Two consequences are used repeatedly below.

Remark 2.6 (The critical gradient). By (2.22), $A(p)$ meets the axis at $p = \frac{1}{2}$ with gradient -4 :

$$A(p) = 2 - 4p + o(1 - 2p) \quad (p \rightarrow \tfrac{1}{2}^-). \quad (2.27)$$

This is the single most useful diagnostic in section 2.7. A candidate closed form may reproduce A to plotting accuracy across the whole interval and still fail here, and failing here is fatal, because as $A \rightarrow 0$ an error in A of vanishing absolute size produces an unbounded error in the quasi-stationary mean $1/A$.

Remark 2.7 (Rate of approach). Numerically, $A(p)/2\varepsilon$ approaches 1 rather slowly. Evaluating (2.13) in high precision gives $A/2\varepsilon = 0.90987, 0.98612, 0.99814, 0.99977$ and 0.999972 at $\varepsilon = 2 \times 10^{-2}$ down to 2×10^{-6} . An exploratory fit to the deficit gives

$$1 - \frac{A}{2\varepsilon} \approx \varepsilon \left(\log \frac{1}{\varepsilon} + c \right), \quad c \approx 0.78, \quad (2.28)$$

consistent with (2.21). The constant c is empirical and is not used anywhere below; the proved statement is the order estimate. What the logarithmic correction does control is where the leading asymptotic becomes an acceptable substitute for the exact value, and hence the crossover chosen in section 2.12.

Why the continuum route is worse

The same leading asymptotic can be reached through the Riccati approximation of the mathematical introduction, and it is worth seeing why that route is inferior. The exact difference is

$$S_{n+1} - S_n = (2p - 1)S_n - pS_n^2 = -\varepsilon S_n - \tfrac{1}{2}S_n^2 + \tfrac{\varepsilon}{2}S_n^2,$$

and dropping the $O(\varepsilon S_n^2)$ term gives $dS/dn = -\varepsilon S - \frac{1}{2}S^2$. Setting $u = 1/S$ linearises this to $u' = \varepsilon u + \frac{1}{2}$, with solution

$$u(n) = Ce^{\varepsilon n} - \frac{1}{2\varepsilon}, \quad S(n) = \frac{1}{Ce^{\varepsilon n} - \frac{1}{2\varepsilon}}.$$

Matching $S_n \sim Ae^{-\varepsilon n}$ for small ε identifies $A = 1/C$, and matching to the critical decay $S_n \sim 2/n$ in the appropriate double limit fixes $C \sim 1/(2\varepsilon)$, recovering (2.22). Both steps are heuristic and the second is delicate; worse, the route gives no control at all on the error, and so cannot see the logarithm of (2.21). The series argument needs no matching and delivers the remainder.

2.6 Why the discrete and continuous constants differ

The two constants are

$$A_c(p) = \frac{1-2p}{1-p} \quad (\text{continuous time}), \quad A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n} \quad (\text{discrete time}),$$

and the observation recorded in the mathematical introduction was that the first starts at 1 where the second starts at $\frac{1}{2}$, yet halving the first does not superpose the curves (figure 2.2). The results of the last two sections settle this completely.



Figure 2.2: The continuous-time constant $A_c(p) = (1-2p)/(1-p)$ (green) against high-precision numerical values of the discrete-time constant $A(p)$ (red points). Both vanish at $p = \frac{1}{2}$, but at $p = 0$ the continuous curve starts at 1 and the discrete one at $\frac{1}{2}$, and the two are not related by a constant factor in between. The black line is a formula obtained by symbolic regression (section 2.7); it tracks the points closely but is not exact.

What holds is not $A = \frac{1}{2}A_c$ but the strict inequality $A > \frac{1}{2}A_c$ of (2.20), and the inequality is attained only in the limit $p \rightarrow 0$. Combining (2.8) and (2.22), the ratio runs monotonically across the interval:

$$\frac{A(p)}{A_c(p)} \rightarrow \frac{1}{2} \quad (p \rightarrow 0), \quad \frac{A(p)}{A_c(p)} \rightarrow 1 \quad (p \rightarrow \frac{1}{2}^-), \quad (2.29)$$

taking the values 0.503, 0.528, 0.619, 0.798, 0.928 and 0.988 at $p = 0.01, 0.1, 0.3, 0.45, 0.49, 0.499$. No constant factor will do, because the ratio is not constant.

The two endpoints have clean interpretations, and they are different in kind.

At $p \rightarrow 0$ the factor of two is a counting effect. In the binary discrete model an individual leaves either nothing or two offspring, so a surviving population is always even (proposition 2.4); deep in the subcritical regime, conditioning on survival means conditioning on a population of exactly two, and the quasi-stationary mean is 2. In the continuous-time model a survivor is a single individual that has not yet died, so the quasi-stationary mean is 1. The reciprocals are $\frac{1}{2}$ and 1, and there is the discrepancy.

At $p \rightarrow \frac{1}{2}^-$ the factor disappears, because both constants are governed by the same critical scaling. Expanding (2.8) near $p = \frac{1}{2}$ gives $A_c(p) = 2\varepsilon/(1 + \varepsilon) \rightarrow 2\varepsilon$, which is precisely (2.22). Near criticality the discreteness of the generational clock stops mattering; what survives is the universal 2ε behaviour, and the two models agree to leading order. They do not agree at the next order: the continuous constant has the elementary expansion $A_c = 2\varepsilon(1 - \varepsilon + O(\varepsilon^2))$, whereas theorem 2.5 shows the discrete correction carries a factor $\log(1/\varepsilon)$ and is therefore asymptotically larger.

The discrete curve is thus squeezed between $\frac{1}{2}A_c$ and its own critical asymptote, hugging the former at one end and the latter at the other. That is the whole content of figure 2.2.

Where the closed form went

It is worth being explicit about the structural difference, because it anticipates section 2.9. In continuous time the survival probability is elementary at every time t , not merely asymptotically:

$$S(t) = \frac{\mu - \lambda}{\mu - \lambda e^{-(\mu - \lambda)t}} \cdot \frac{\mu}{\mu} \quad \text{rearranges to} \quad S(t) = \frac{\mu - \lambda}{\mu} e^{-(\mu - \lambda)t} \left(1 - \frac{\lambda}{\mu} e^{-(\mu - \lambda)t}\right)^{-1}, \quad (2.30)$$

so the amplitude $(\mu - \lambda)/\mu$ can simply be read off the exact solution. In discrete time no such exact solution exists: the survival probability is defined by iterating a quadratic map, and its amplitude is the infinite product (2.13) of every early correction. The continuous process is solvable because its generating function satisfies a linear first-order partial differential equation with an elementary characteristic; the discrete one is not because iterating a quadratic map is, in the precise sense of section 2.9, a transcendental operation. The factor of $\frac{1}{2}$ at $p = 0$ is a parity accident; the absence of a closed form is not.

2.7 Searching for a closed form

Before the connection of section 2.8 was appreciated, several methods were tried in the hope of an elementary formula for $A(p)$. The negative result is worth recording, and so is the manner of the failure, because the manner of the failure is itself evidence.

The first method was the continuum route of section 2.5: treat the map as a differential equation in the large-generation limit and solve it. This gives the shape of the decay but leaves the constant of integration undetermined, which is to say it leaves $A(p)$ undetermined. That is not an accident of the method. The constant of integration is fixed by the early generations, where the continuum approximation is invalid, and $A(p)$ is exactly the accumulated record of those generations.

The second was symbolic regression by genetic algorithm. The idea is simple: generate fifty candidate functions at random, measure how closely each reproduces the target curve, retain the best five, and build a fresh population of fifty by varying those; then measure again, and repeat the cycle of variation and selection many times over [1, 2]. The first implementation searched a space of polynomials, on the hunch that the answer might take that form and because such functions are simple to encode: a genome is a coefficient list $[a_0, \dots, a_k]$ for $a_0 + a_1p + \dots + a_kp^k$ up to some maximum order, drawn uniformly at random. It became apparent quickly that many substantially different formulae produce curves indistinguishable from the target by eye. Supposing that a true formula might have rational coefficients, a second version searched $\sum_i (a_i/b_i)p^i$ with $[a_i], [b_i] \in \mathbb{Z}^{k+1}$, and a third extended the space to ratios of two such polynomials. Candidates could be constrained to the known endpoints $A(0) = \frac{1}{2}$ and $A(\frac{1}{2}) = 0$, and the best that emerged fitted the high-precision

numerical data well:

$$\hat{A}_1(p) = \frac{\frac{1}{2}(1-2p)(1+2p)}{1 + \frac{1}{2}p - \frac{5}{2}p^2 - \frac{6}{5}p^3 + \frac{6}{5}p^4}. \quad (2.31)$$

This passes through $(0, \frac{1}{2})$ and $(\frac{1}{2}, 0)$ and looks convincing, but it fails the diagnostic of [remark 2.6](#): differentiating at $p = \frac{1}{2}$ gives

$$\hat{A}'_1(\tfrac{1}{2}) = -\frac{2}{0.55} \approx -3.636$$

rather than the required -4 . Matching the endpoint values while missing the slope is enough to make the relative error in A , and hence the error in $1/A$, unacceptable near criticality.

A later attempt used a symbolic tree search over progressively nested elementary functions of all kinds. The conclusion was the same. Many functions of widely varying form resemble the target closely, and no satisfactory simple formula emerges even when candidates with fewer components are explicitly favoured. The expressions returned were typically deep nestings of trigonometric and iterated exponential terms carrying a dozen or more fitted decimal constants, and they were not interpretable in any useful sense; a representative and comparatively concise one is

$$\hat{A}_3(p) = \frac{-0.616647881739505 e^{-0.552605335919693 e^p}}{\sin(\cos p) - |p|} + 0.921964323679771. \quad (2.32)$$

These expressions have no inferential status. They are records of a curve-fitting exercise, not proposed identities, and they are reproduced here only to show what the search returned.

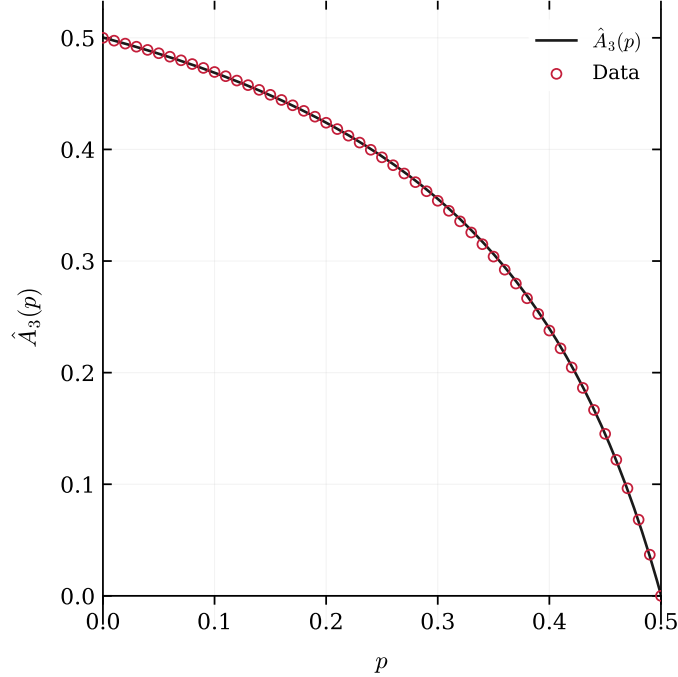


Figure 2.3: The symbolic-regression candidate $\hat{A}_3(p)$ of (2.32) (solid) against numerically computed values of $A(p)$ (red circles). The agreement is excellent to the eye across the whole interval, which is exactly the problem: the fit is not evidence of correctness. $\hat{A}_3$ misses the critical point, returning $\hat{A}_3(\frac{1}{2}) = 9.1 \times 10^{-4}$ rather than 0, and arrives there with gradient -3.63 instead of -4 . Errors of this size are invisible here but are fatal for $1/A$ near criticality.

The recurring defect is the same in every case. The candidates either fail to pass through $(\frac{1}{2}, 0)$ or fail to arrive there with gradient -4 , and near criticality this matters enormously: as $A \rightarrow 0$, an error in A of vanishing absolute size produces an $O(1)$ — eventually unbounded — error in the quasi-stationary mean $1/A$. A formula can look perfect on a plot of A and be useless on a plot of $1/A$.

A pragmatic patch is to splice the asymptotic onto a fitted curve,

$$\hat{A}_4(p) = \begin{cases} -0.04011483983619545 e^{e^{2p}} + 0.6079994336528085, & p < 0.499962, \\ 2 - 4p, & p \geq 0.499962, \end{cases} \quad (2.33)$$

and this does work as a surrogate. But if one is going to splice, it is better to splice onto something exact: the certified partial product (2.15) on the first branch and the proved asymptotic (2.21) on the second. Section 2.12 does exactly that.

The deeper lesson points directly at the next section. That an enormous number of very different formulae can behave almost identically over a bounded interval is not merely an inconvenience of the method; it is a symptom. Curve fitting cannot distinguish a function that has a closed form from one that does not, and no quantity of fitting will ever settle the question. Something structural is needed, and there is something structural available.

2.8 The Koenigs identification

Some time after the search of section 2.7 had been abandoned, a dynamical argument emerged that explains why no elementary closed form should be expected. In short: $A(p)$ is a value of the Koenigs linearising function of the logistic map, and that function is known to be non-elementary throughout the parameter range required. This section establishes the identification; the next proves the non-elementarity.

2.8.1 From the survival recursion to the logistic map

The substitution has already been made in section 2.3. With $S_n = 2w_n$ and $r = 2p$, the survival recursion (2.2) becomes the logistic iteration

$$w_{n+1} = f_r(w_n), \quad f_r(w) = r w(1 - w), \quad w_0 = \frac{1}{2}, \quad (2.34)$$

with $r \in [0, 1)$ throughout the subcritical regime, and

$$A(p) = 2 \lim_{n \rightarrow \infty} \frac{w_n}{r^n}. \quad (2.35)$$

So the constant is an asymptotic invariant of a quadratic map iterated from a specific point — which is exactly the object classical complex dynamics was built to describe.

2.8.2 Koenigs linearisation

Definition 2.8 (Koenigs function). Let $f_r(z) = rz(1 - z)$ have a fixed point at $z = 0$ with multiplier $r = f'_r(0)$ satisfying $0 < |r| < 1$. Koenigs [3] proved that there is a unique function ψ_r , holomorphic in a neighbourhood of the origin, satisfying the Schröder equation

$$\psi_r(f_r(z)) = r \psi_r(z), \quad \psi_r(0) = 0, \quad \psi'_r(0) = 1. \quad (2.36)$$

Equivalently,

$$\psi_r(z) = \lim_{n \rightarrow \infty} r^{-n} f_r^{\circ n}(z) \quad (2.37)$$

uniformly near the origin. The normalisation $\psi'_r(0) = 1$ is equivalent to $\psi_r(z)/z \rightarrow 1$ as $z \rightarrow 0$.

The Koenigs function is a change of coordinate that straightens the dynamics: in the ψ coordinate the nonlinear map f_r becomes multiplication by r (figure 2.4). Here f_r is exactly (2.34), with $z = w_n$ and $f_r(z) = w_{n+1}$, and the multiplier is $r = 2p \in (0, 1)$ — an attracting fixed point, which is the classical Koenigs setting. Standard accounts are Milnor [4] and Carleson and Gamelin [5].

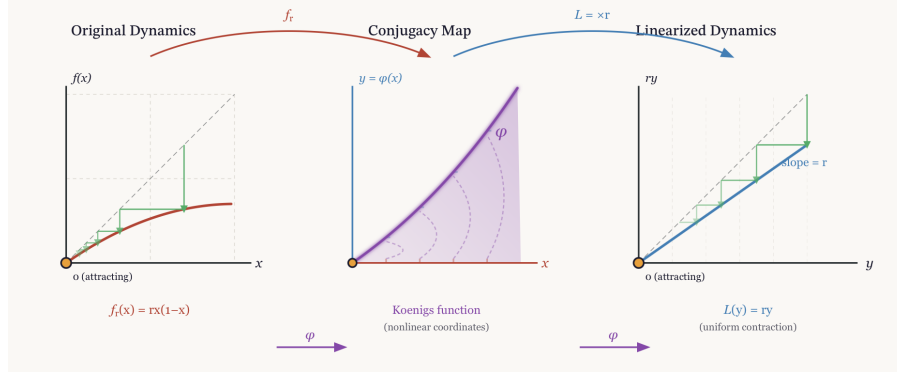


Figure 2.4: Koenigs linearisation of the logistic map near an attracting fixed point. The nonlinear dynamics $x \mapsto f_r(x) = rx(1-x)$ are conjugated by ψ_r to the linear map $y \mapsto ry$; the diagram displays the functional relation $\psi_r(f_r(x)) = r\psi_r(x)$ and the role of ψ_r as the coordinate change that straightens the dynamics.

2.8.3 The identity

One point of care is required before the identification can be stated. Koenigs' theorem is local: the series (2.37) is guaranteed to converge only on a disc about the origin, and for the logistic map that disc has radius $(1-r)/r$. The evaluation point of interest is $w_0 = \frac{1}{2}$, and $\frac{1}{2} < (1-r)/r$ fails as soon as $r \geq \frac{2}{3}$, that is $p \geq \frac{1}{3}$. For those parameters $\frac{1}{2}$ lies outside the germ, and writing " $\psi_r(\frac{1}{2})$ " as a naive evaluation of a power series would be meaningless.

The remedy is the functional equation itself. For $r \in (0, 1)$ the origin is attracting on the real line and the whole interval $(0, 1)$ lies in its real basin, so the orbit $w_n = f_r^{\circ n}(w_0)$ enters the germ after finitely many steps. Define the *basin extension*

$$\Psi_r(w) := \lim_{n \rightarrow \infty} r^{-n} f_r^{\circ n}(w), \quad w \in (0, 1), \quad (2.38)$$

equivalently $\Psi_r(w) = r^{-N} \psi_r(f_r^{\circ N}(w))$ for any N large enough that $f_r^{\circ N}(w)$ lies in the germ. The two definitions agree where both apply, because (2.36) makes the right-hand side independent of N , and Ψ_r is the unique analytic continuation of ψ_r along the orbit.

Proposition 2.9 (The constant as a Koenigs value). *For $p \in (0, \frac{1}{2})$ and $r = 2p$,*

$$A(p) = 2 \Psi_r\left(\frac{1}{2}\right) = 2 \lim_{n \rightarrow \infty} r^{-n} f_r^{\circ n}\left(\frac{1}{2}\right). \quad (2.39)$$

Proof. Iterating (2.36) along the orbit $w_0, w_1, w_2, \dots$ of (2.34), from any point at which the germ applies, gives $\psi_r(w_n) = r \psi_r(w_{n-1})$ and hence

$$\Psi_r(w_0) = \frac{\psi_r(w_n)}{r^n} \quad \text{for every } n \text{ large enough that } w_n \text{ lies in the germ.} \quad (2.40)$$

Passing to the limit, $\Psi_r(w_0) = \lim_n \psi_r(w_n)/r^n$. The normalisation $\psi_r(0) = 0$, $\psi'_r(0) = 1$ means $\psi_r(z)/z \rightarrow 1$ as $z \rightarrow 0$, and $w_n = S_n/2 \rightarrow 0$ because $r < 1$; replacing $\psi_r(w_n)$ by w_n in the limit is therefore legitimate, and $\Psi_r(w_0) = \lim_n w_n/r^n$. Comparing with (2.35) and using $w_0 = S_0/2 = \frac{1}{2}$ gives $A(p) = 2\Psi_r(\frac{1}{2})$. $\square$

We write $A(p) = 2\psi_r(\frac{1}{2})$ below as shorthand for this basin value, never as a naive evaluation of a power series beyond its guaranteed disc of convergence. Every question about an elementary closed form for $A(p)$ is thereby tied to the nature of the conjugacy at multiplier $r = 2p$.

Remark 2.10 (Germ versus basin). The distinction just drawn will matter again in section 2.9, so it is worth recording what it does *not* cost. Differential algebraicity is a local condition preserved under analytic continuation: if an algebraic differential equation held for ψ_r on any subdomain, it would persist along the continuation defining Ψ_r , and conversely. Hypertranscendence of the germ and of the basin extension are therefore equivalent statements, and the identity (2.39) is unaffected by which one is under discussion.

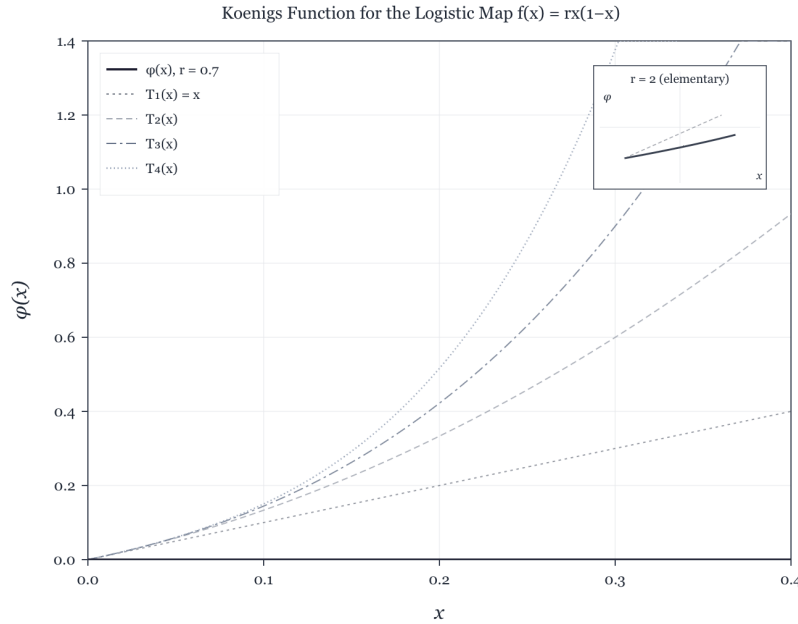


Figure 2.5: Numerical approximation of a Koenigs function for a logistic multiplier in $(0, 1)$, with low-order Taylor polynomials near the origin. Elementary closed forms exist only for exceptional multipliers outside this range; those cases are derived in section 2.A.

2.9 No elementary conjugacy in the subcritical window

This section makes the obstruction precise. The logical skeleton is short:

- (i) every elementary function satisfies an algebraic differential equation;
- (ii) differential algebraicity survives compositional inversion;
- (iii) by a theorem of Becker and Bergweiler, a solution of Schröder's equation for a rational map of degree at least two can be differentially algebraic only when the underlying fixed point is *repelling*;
- (iv) for $r \in (0, 1)$ the fixed point 0 of f_r is attracting.

Hypertranscendence of ψ_r at every subcritical parameter follows at once — with no exceptional cases to exclude, because the attracting window contains no candidates at all. [Section 2.10](#) then explains, with equal care, what this does *not* settle.

2.9.1 Differential algebraicity, hypertranscendence, elementarity

Definition 2.11 (Differentially algebraic; hypertranscendental). A function g , holomorphic on a domain, or a formal power series, is *differentially algebraic over $\mathbb{C}(z)$* if there is a non-zero polynomial P in $n + 2$ variables, for some $n \geq 0$, with

$$P(z, g(z), g'(z), \dots, g^{(n)}(z)) \equiv 0. \quad (2.41)$$

Otherwise g is *differentially transcendental*, or *hypertranscendental*.

Two closure facts make this the right notion for the present purpose.

Lemma 2.12 (Elementary implies differentially algebraic). *Every elementary function in the sense of Liouville — any function reachable from $\mathbb{C}(z)$ by a finite tower of algebraic operations, exponentials and logarithms — is differentially algebraic. Consequently a hypertranscendental function is not elementary.*

Proof sketch. Each storey of a Liouvillian tower preserves differential algebraicity: algebraic extensions do so by elimination; if g is differentially algebraic then so are e^g and $\log g$, their defining first-order relations combining with the equation for g ; and the class is closed under the field operations and under composition. These closure properties are classical; see Rubel's survey [6]. $\square$

Lemma 2.13 (Closure under compositional inversion). *Let g be holomorphic near 0 with $g(0) = 0$ and $g'(0) \neq 0$, and let g^{-1} denote its local compositional inverse. Then g is differentially algebraic over $\mathbb{C}(z)$ if and only if g^{-1} is (Moore [7]; see also Boshernitzan and Rubel [8]).*

[Lemma 2.13](#) matters because the literature states Schröder-equation results in two equivalent but opposite conventions, and conflating them is an easy way to misquote a theorem. Our ψ_r is the *linearising coordinate*: $\psi_r(f_r(z)) = r\psi_r(z)$. The function classified by Becker and Bergweiler — and by Ritt before them — is the *linearising parametrisation* $\varphi_r = \psi_r^{-1}$, which satisfies

$$\varphi_r(rt) = f_r(\varphi_r(t)), \quad \varphi_r(0) = 0, \quad \varphi'_r(0) = 1. \quad (2.42)$$

By [lemma 2.13](#), hypertranscendence of either function transfers to the other.

2.9.2 The Becker–Bergweiler theorem, stated precisely

The result needed is the 1995 classification of differentially algebraic Schröder functions [9]; we quote it in the form given in Fernandes’ survey of the Schröder, Böttcher and Abel equations [10, Thm 3.1].

Theorem 2.14 (Becker–Bergweiler [9], as restated in [10, Thm 3.1]). *Let R be a rational map of degree at least two with $R(0) = 0$ and multiplier $s = R'(0)$, where $s \neq 0$ and s is not a root of unity; for $|s| = 1$ the Schröder function is understood as a formal series. Let φ , normalised by $\varphi(0) = 0$ and $\varphi'(0) = 1$, solve $\varphi(st) = R(\varphi(t))$. If φ is differentially algebraic over $\mathbb{C}(t)$, then*

- (i) *the fixed point 0 is repelling, $|s| > 1$; and*
- (ii) *φ is a Möbius transformation of one of*

$$\exp(\alpha t^q), \quad \cos(\alpha t^q + \beta), \quad \wp(\alpha t^q + \beta), \quad \wp^2, \quad \wp^3, \quad \wp'(\alpha t^q + \beta),$$

where $\wp$ is a Weierstrass function, $q \in \mathbb{Q}$ is such that the function concerned is meromorphic on $\mathbb{C}$, $\alpha \neq 0$, and β is a fraction of a period. Correspondingly R is Möbius-conjugate to $z^{\pm d}$, to $\pm T_d$ (Chebyshev), or to a Lattès map.

Remark 2.15 (Lineage and independent corroboration). For $|s| > 1$ — the Poincaré-function case — the classification goes back to Ritt in 1926 [11]; Becker and Bergweiler completed the picture for the three classical functional equations at once, and Bergweiler’s later account with Aschenbrenner [12] assigns the exceptional list explicitly to the repelling case. Di Vizio and Fernandes [13] have given an independent difference-Galois proof of Ritt’s theorem; notably, their Theorems 1.2 and 4.2 assume only that s is neither zero nor a root of unity — no condition $|s| > 1$ — and show that any differentially algebraic solution of $y(R(x)) = sy(x)$, which is our ψ -convention with no inversion needed, must already satisfy a first-order equation of Ritt type $(y')^e = B(x)y^j$. The structure underlying theorem 2.14 therefore does not hang on a single paper.

2.9.3 Hypertranscendence throughout the subcritical window

Theorem 2.16 (No exceptional parameters in $(0, 1)$). *For every $r \in (0, 1)$, the Koenigs function ψ_r of $f_r(z) = rz(1 - z)$ at 0 is hypertranscendental over $\mathbb{C}(z)$: it satisfies no algebraic differential equation. In particular ψ_r is not an elementary function.*

Proof. Fix $r \in (0, 1)$. The map f_r is rational of degree two, fixes 0, and its multiplier $s = f'_r(0) = r$ satisfies $0 < s < 1$; the hypotheses of theorem 2.14 hold, with 0 attracting. Suppose for contradiction that ψ_r were differentially algebraic. Its local inverse $\varphi_r = \psi_r^{-1}$ solves the parametrisation form (2.42) with the stated normalisation, and by lemma 2.13 it would be differentially algebraic too. Theorem 2.14(i) would then force $|r| > 1$, contradicting $r \in (0, 1)$. Hence ψ_r is hypertranscendental, and non-elementarity follows from lemma 2.12. $\square$

Remark 2.17 (The exceptional set inside the window is empty, not merely avoided). An earlier form of this argument reasoned through the parameter plane. The affine conjugacy $c(r) = (2r - r^2)/4$ of lemma 2.18 sends $r \in (0, 1)$ to $c \in (0, \frac{1}{4})$, which avoids the exceptional quadratic parameters $c = 0$ (reached at $r = 2$) and $c = -2$ (reached at $r = 4$, and also at $r = -2$). That observation is true but logically superfluous, and it invites a question it cannot answer: *is the exceptional list, restricted to the window, complete?* The route through theorem 2.14(i) dissolves the question. The exceptional Schröder functions exist only over *repelling* fixed points, so the attracting window contains no

candidates whatsoever, and no enumeration of exceptional parameters is required. Consistently, the classical elementary cases of [section 2.A](#) have multipliers 2 and 4 at the linearised fixed point — both repelling: they are Poincaré-type linearisations, and neither can be pressed into service for $r \in (0, 1)$.

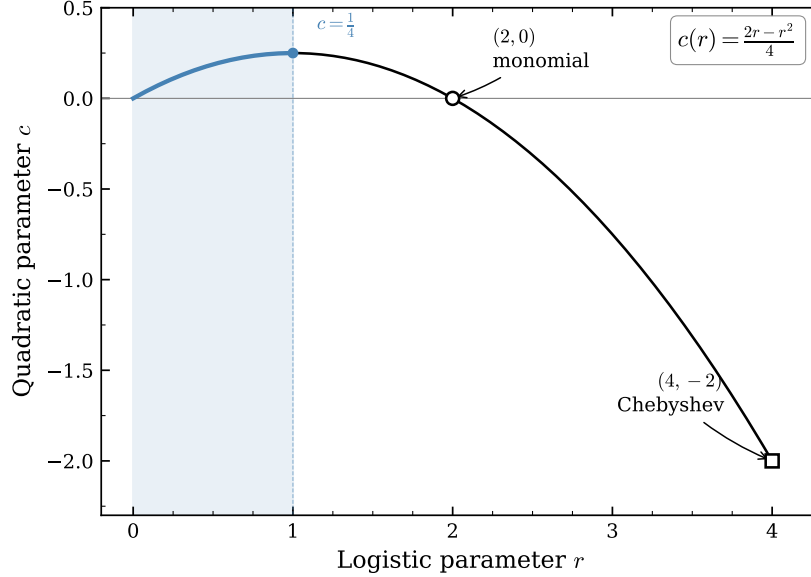


Figure 2.6: The affine parameter correspondence $c(r) = (2r - r^2)/4$ between the logistic family $f_r(x) = rx(1 - x)$ and the quadratic family $P_c(z) = z^2 + c$. The shaded segment is $r \in (0, 1)$, the subcritical branching window, which maps to $c \in (0, \frac{1}{4})$; the two integrable parameters, $r = 2$ ($c = 0$, monomial) and $r = 4$ ($c = -2$, Chebyshev), lie outside it. In the light of [remark 2.17](#) this picture is a consistency check rather than the proof: the exceptional linearisations live over repelling fixed points, which the attracting window excludes outright.

2.9.4 Where the window sits in the quadratic family

It is worth seeing where in the quadratic family these parameters sit, because the picture makes the difficulty look structural rather than accidental — though it is, again, illustration and not argument. The main cardioid of the Mandelbrot set — the set of c for which $z^2 + c$ has an attracting fixed point — is parametrised by that point's multiplier λ through $c = \lambda/2 - \lambda^2/4$, which is exactly the correspondence of [lemma 2.18](#) with $\lambda = r$. The branching window $r \in (0, 1)$ is therefore the real diameter of the main cardioid, traversed from the centre $c = 0$ out to the parabolic cusp $c = \frac{1}{4}$, where $\lambda = 1$ and the process turns critical. [Figure 2.7](#) places the segment in context.

Throughout the window the Julia set is a quasicircle bounding the basin of the attracting fixed point, and it is a genuine fractal for every c in the interval. The Koenigs function is the conformal coordinate that linearises the dynamics inside that basin, so its natural domain is bounded by exactly this curve. A function whose domain of definition has no smooth description is not a promising candidate for a closed form, and $A(p)$, being one of its values, inherits the difficulty. The contrast with the integrable cases is sharp: at $c = 0$ the Julia set degenerates to the unit circle, and it is precisely there that the linearisation becomes elementary, [\(2.45\)](#). None of this is used as evidence for the theorem, which [section 2.9.3](#) has already proved by a route that never enters the parameter plane.

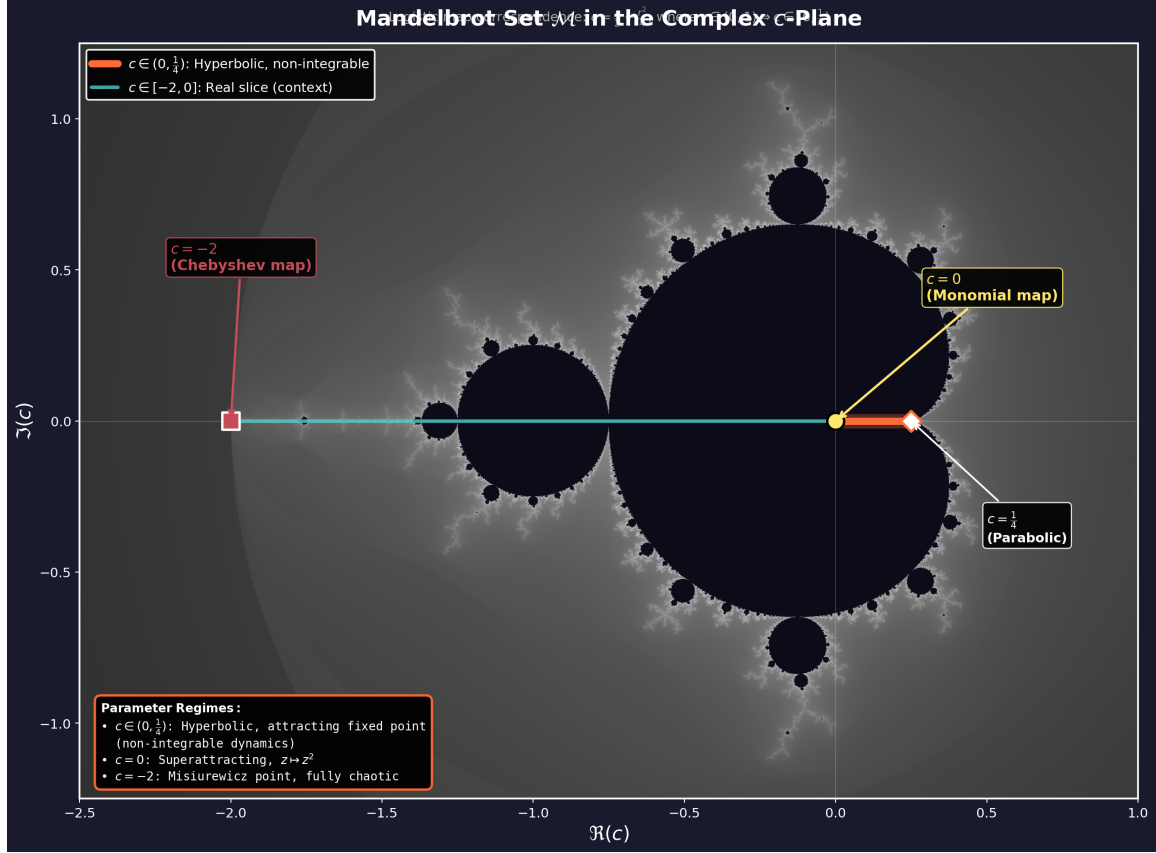


Figure 2.7: The Mandelbrot set in the complex c -plane, with the interval $c \in (0, \frac{1}{4})$ highlighted. Under the correspondence $c = (2r - r^2)/4$ of lemma 2.18 this is the image of the subcritical branching window $r = 2p \in (0, 1)$, and it forms the real diameter of the main cardioid, running from the centre $c = 0$ to the parabolic cusp $c = \frac{1}{4}$, which is criticality. The two integrable parameters are marked: $c = 0$ (monomial, reached in logistic coordinates at $r = 2$) and $c = -2$ (Chebyshev, $r = 4$), the latter at the far tip of the real slice. Note that $c(r) = c(2 - r)$, so each $c \in (0, \frac{1}{4})$ is the image of two logistic parameters, r and $2 - r$; these correspond to the two fixed points of $z^2 + c$, with multipliers r and $2 - r$, and it is the attracting one, $r < 1$, that governs $A(p)$. The figure is a pedagogical consistency check, not part of the proof of theorem 2.16.

2.10 What is and is not proved about $A(p)$

“Closed form” must be made precise before any claim is stated, because at least four inequivalent readings are in play and a result for one is not a result for another:

- (V) the *value* $A(p_0)$ is elementary, or algebraic, for a fixed rational p_0 ;
- (E) the *map* $p \mapsto A(p)$ is an elementary function of p ;
- (D) the map is *differentially algebraic* in p , that is, it satisfies some algebraic ordinary differential equation in p ;
- (S) the map is expressible through standard special functions (Γ , ζ , ${}_2F_1$, theta or q -series).

Theorem 2.16 speaks to the function $z \mapsto \psi_r(z)$ at fixed r , and licenses exactly this much: *no*

evaluation of $A(p)$ can be short-circuited through an algebraic differential equation for the lineariser, at any subcritical parameter. It does not, by itself, decide any of (V), (E), (D) or (S). Two gaps must be stated plainly.

The value gap

A hypertranscendental function can take elementary — even algebraic — values at individual points: Γ is hypertranscendental by Hölder’s theorem, and yet $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Nothing established so far forbids, say, $A(\frac{1}{4})$ from being an algebraic number.

There *is* a genuine values theorem in this circle of ideas. Becker and Bergweiler proved in 1993 [14] that the Böttcher conjugacy between two polynomials of the *same degree* $d \geq 2$ with algebraic coefficients, not linearly conjugate to monomials or Chebyshev polynomials, is transcendental and takes transcendental values at algebraic points; Mahler’s method drives the proof (for the method see Nishioka [15]). One might hope to transfer this to the Koenigs value $\psi_r(\frac{1}{2})$. It does not transfer, for two independent reasons.

Formally, the Schröder equation pairs an inner map of degree two with an outer map — multiplication by r — of degree one, so the equal-degree hypothesis fails outright.

Structurally, no easy repair exists. Mahler’s method balances doubly-exponential analytic decay along an orbit against comparable height growth, and in the Böttcher setting both exponents run like d^n . In the attracting Koenigs setting the orbit $w_n = f_r^{\circ n}(\frac{1}{2})$ decays only geometrically, $|w_n| \sim A r^n$, while the heights of the rational numbers w_n still grow doubly exponentially, $h(w_n) \sim C 2^n$; the resulting inequalities run the wrong way at every n . Note that the obstruction is *not* the germ/basin issue of [remark 2.10](#) — the pullback there moves the evaluation point arbitrarily deep into the germ at the cost of a rational factor — it is the degree structure of the equation itself.

The consequence deserves emphasis:

*transcendence — indeed, even irrationality — of $A(p_0)$ is at present not proved
for a single rational $p_0 \in (0, \frac{1}{2})$.*

The parameter gap

[Theorem 2.16](#) says nothing about the map $p \mapsto A(p)$: readings (E) and (D) are open. To the best of our knowledge the question is not merely unsolved but unaddressed in the literature. The natural machinery is the parametrised differential Galois theory of difference equations of Hardouin and Singer [16], designed precisely to detect differential relations in auxiliary directions for linear difference equations such as $\sigma\psi = r\psi$; it does not apply as it stands, because the derivation ∂_r fails to commute with the substitution operator $\sigma: w \mapsto f_r(w)$, which itself moves with r . Nor does any feature of the near-critical asymptotics obstruct (E) or (D): the logarithmic corrections to (2.21) are themselves elementary functions of ε .

2.11 Inverse-symbolic evidence at eleven rational parameters

Since the theorems leave (V) open, the value question was attacked directly, by integer-relation detection. The values

$$A(p), \quad p \in \left\{ \frac{1}{10}, \frac{1}{8}, \frac{1}{6}, \frac{1}{5}, \frac{1}{4}, \frac{3}{10}, \frac{1}{3}, \frac{3}{8}, \frac{2}{5}, \frac{5}{12}, \frac{9}{20} \right\}, \quad (2.43)$$

were computed to more than 1150 significant digits by two mechanically independent routes — the exact product identity (2.13) iterated to convergence, and a pullback of the orbit into the Koenigs germ followed by evaluation of the Koenigs series — which agreed to at least 1192 digits at every parameter. Each value was then subjected to a PSLQ battery at ≥ 200 -digit effective rigour:

- *algebraicity*: no integer polynomial relation of degree ≤ 10 and coefficient height $\leq 10^6$;
- *linearity*: no rational linear relation, at height $\leq 10^8$, against the constants π , π^2 , $\ln 2$, $\ln 3$, $\ln 5$, γ , $\zeta(3)$, $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, e , e^π , $\Gamma(\frac{1}{3})$, $\Gamma(\frac{1}{4})$ and Catalan’s constant — also tested for $1/A$, the quasi-stationary mean itself, and for A/ε ;
- *bilinearity*: no representation $A = L_1/L_2$ with L_1, L_2 integer linear forms of height $\leq 10^8$ in a twelve-constant basis, run at 460 digits;
- *multiplicativity*: no relation $A = e^{q_0} 2^{q_1} 3^{q_2} 5^{q_3} 7^{q_4} \pi^{q_5}$ with rational exponents of height $\leq 10^{10}$;
- *cross-parameter*: no rational linear or multiplicative relation linking the eleven values to one another, run at 500 digits.

In all, 123 tests returned no relation — indeed no discovery-stage candidate: every PSLQ run terminated on its internal norm bound, so each null carries a genuine “no relation up to the stated height” guarantee. Throughout, the precision head-room, digits divided by basis size, exceeded every height cap by at least eight orders of magnitude, so a returned relation would have been meaningful rather than numerical noise.¹

This is finite evidence, and its scope must be stated honestly: it excludes low-complexity closed forms over the tested constants, and nothing more. A closed form over untested constants — $\Gamma(\frac{1}{7})$, say, or periods of higher-genus curves — or one of height above the caps would have escaped the search. It is evidence for the conjecture, not a proof of it.

2.11.1 The working claim

For every fixed $r = 2p \in (0, 1)$ the Koenigs linearisation ψ_r is hypertranscendental in its own variable (theorem 2.16) — a theorem, with no exceptional parameters in the attracting window, since differentially algebraic Schröder functions exist only over repelling fixed points. Consequently no elementary expression in z for the conjugacy can be evaluated to produce $A(p)$. Whether individual values $A(p_0)$ are transcendental, or even irrational, and whether the map $p \mapsto A(p)$ is elementary or differentially algebraic in p , remain open questions. No closed form under any reading is known, none is expected, and eleven rational values resist inverse-symbolic identification against standard constants at over 200-digit rigour.

Full derivations of the elementary solutions at $r = 2$ and $r = 4$, and their place inside the Becker–Bergweiler classification, are collected in section 2.A.

2.12 Computing $A(p)$ in practice

In the absence of an exact formula we are left with approximation, and the two regimes identified above suggest how to split the work.

¹Computations in `mpmath` at working precisions of 310–1200 digits; the values, scripts and full per-test log are archived with the project sources and are not reproduced in this thesis.

Over most of the interval, direct iteration is entirely satisfactory: the infinite product (2.13) converges quickly, its partial products bound $A(p)$ from above at every order, the tail bound (2.15) supplies a matching lower bound and hence a stopping certificate, and modest numbers of iterations give many digits. The difficulty is confined to a neighbourhood of $p = \frac{1}{2}$, and it has two parts. First, as $A(p) \rightarrow 0$ the quantity of interest $1/A$ diverges, so a fixed absolute error in A becomes an unbounded error in the quasi-stationary mean. Second, and this is the binding constraint, as the process is defined ever closer to criticality its lifetime, though still finite, grows enormously, and S_n recedes to zero ever more slowly. The iteration must be carried to a generation at which S_n is essentially unchanging, and table 2.1 shows what that costs: about 10^2 iterations at $p = 0.2$, 10^3 at $p = 0.45$, but nearly 5×10^5 at $p = 0.4999$.

The remedy is to hand the near-critical region to the asymptotic (2.21), which is exact in the limit and cheap everywhere:

$$A(p) \approx \begin{cases} \frac{1}{2} \prod_{k=1}^N \left(1 - \frac{S_k}{2}\right), & p < p_*, \quad N \gg 1, \\ 2 - 4p, & p \geq p_*, \end{cases} \quad (2.44)$$

with the crossover p_* chosen just below $\frac{1}{2}$. Remark 2.7 indicates where to put it. The relative error of the asymptotic is about $\varepsilon \log(1/\varepsilon)$ by theorem 2.5, so at $p_* = 0.4999$ (that is $\varepsilon = 2 \times 10^{-4}$) the asymptotic is already accurate to about two parts in 10^3 , while the product still converges in a few hundred thousand iterations. Any p_* in the range 0.499 to 0.4999 gives a scheme accurate to better than one per cent throughout, and the two branches can be blended over a narrow window if continuity of the derivative matters.

Three practical points are worth recording.

First, use the product rather than the raw ratio $S_n/(2p)^n$. Both require the same iteration, but the product is monotone decreasing in N and comes with the two-sided certificate (2.15), whereas the ratio approaches its limit from no fixed side and offers no error control.

Second, stop on the certificate, not on apparent convergence. In the near-critical regime successive floating-point iterates cease to differ long before the tail is genuinely negligible, and a stopping rule based on observed change silently returns the wrong answer. The bracket in (2.15) costs one extra multiplication and removes the failure mode.

Third, if the quantity actually needed is the quasi-stationary mean $1/A$ rather than A itself — as it usually is — then the series (2.17) computes it directly, and its partial sums increase to the answer, giving the complementary one-sided bound.

This is the recipe used for the numerical values quoted throughout this chapter and in the mathematical introduction, including table 2.1.

2.13 Conclusion

The constant $A(p)$ organises the quasi-stationary behaviour of subcritical binary Galton–Watson branching: the limiting conditional mean is $1/A(p)$, the limiting conditional variance is (2.7), and both are functions of p and $A(p)$ alone. This chapter has shown that the constant is well defined (proposition 2.1), exactly representable as an infinite product (2.13) and as a series (2.17), confined by proposition 2.3 between $\frac{1}{2}A_c(p)$ and $2\varepsilon/(1+2\varepsilon)$ and by proposition 2.4 below $\frac{1}{2}$, asymptotically equal to 2ε near criticality with a relative correction of order $\varepsilon \log(1/\varepsilon)$ (theorem 2.5), and computable to arbitrary accuracy with a certificate by the hybrid scheme (2.44). For every purpose this thesis has, $A(p)$ is a usable object.

What it is not is elementary, and the reason is structural rather than a failure of ingenuity. By [proposition 2.9](#) the constant is twice the basin value at $\frac{1}{2}$ of the Koenigs linearising function of the logistic map at multiplier $r = 2p$, and by [theorem 2.16](#) that function is hypertranscendental in its own variable for every r in the subcritical window — with no exceptional parameters to exclude, since differentially algebraic Schröder functions exist only over repelling fixed points, and the window is entirely attracting. This explains the failure of every closed-form search recorded in [section 2.7](#), and it explains the characteristic form of that failure: an object defined only by the dynamics that generate it will be approximated well by many unrelated expressions and captured by none.

What the theorem does not decide has been flagged with equal care in [section 2.10](#). Transcendence — even irrationality — of the individual values $A(p_0)$ is open, because the 1993 Becker–Bergweiler values theorem is confined to equal-degree Böttcher pairs and does not transfer, the degree structure of the Schröder equation being wrong and Mahler’s method running the wrong way in the attracting case. Whether the map $p \mapsto A(p)$ is elementary, or differentially algebraic in p , is open too, and appears to be unaddressed: the obvious machinery of parametrised difference Galois theory does not apply because the parameter derivation does not commute with a substitution operator that itself moves with the parameter. Eleven rational values nonetheless resist all inverse-symbolic identification against standard constant bases at over 200-digit rigour, which is evidence for the conjecture that no closed form exists under any reading, and is not a proof of it.

Two problems are therefore left open by this chapter, and they are worth separating from the modelling questions raised in the mathematical introduction.

1. *The value problem.* Is $A(p_0)$ irrational for some, or every, rational $p_0 \in (0, \frac{1}{2})$? A positive answer for a single parameter would be the first arithmetic result about this constant. The PSLQ battery of [section 2.11](#) bounds the difficulty from one side without touching the question.
2. *The parameter problem.* Is $p \mapsto A(p)$ differentially algebraic in p ? This is the reading under which a “formula for $A(p)$ ” would most naturally be sought, and it is precisely the reading [theorem 2.16](#) does not address. A negative answer would close the matter; a positive one would be more surprising still.

The exceptional elementary linearisations at $r = 2$ and $r = 4$, both over repelling fixed points and both outside the branching window, are derived in [section 2.A](#) and located inside the Becker–Bergweiler classification there. They are recorded to show what a solvable case looks like, not because they bear on $A(p)$.

2.A Elementary Koenigs functions at $r = 2$ and $r = 4$

These cases lie outside subcritical branching, where $r = 2p < 1$, but they complete the classical picture of integrable logistic linearisations and they show what a solvable case looks like. At both parameters the fixed point at the origin is *repelling*, so the functions below are Poincaré-type linearisations rather than attracting Koenigs maps; [remark 2.17](#) explains why that is the decisive point rather than an incidental one.

2.A.1 The monomial case, $r = 2$

For $r = 2$ we have $f_2(x) = 2x(1 - x)$, and the fixed point 0 has multiplier $f'_2(0) = 2$. The affine change $u = 1 - 2x$ conjugates the dynamics to the monomial map: since

$$1 - 2f_2(x) = 1 - 4x + 4x^2 = (1 - 2x)^2,$$

the substitution gives $u_{n+1} = u_n^2$, which the logarithm linearises because $\ln(u^2) = 2 \ln u$. Returning to x and imposing the normalisation of [definition 2.8](#),

$$\psi_2(x) = -\frac{1}{2} \ln(1 - 2x), \quad (2.45)$$

which is readily verified:

$$\psi_2(f_2(x)) = -\frac{1}{2} \ln((1 - 2x)^2) = -\ln(1 - 2x) = 2\psi_2(x),$$

with $\psi_2(0) = 0$ and $\psi'_2(0) = 1$.

2.A.2 The Chebyshev case, $r = 4$

For $r = 4$ we have $f_4(x) = 4x(1 - x)$, with multiplier $f'_4(0) = 4$. Putting $x = \sin^2 \theta$,

$$f_4(x) = 4 \sin^2 \theta \cos^2 \theta = (2 \sin \theta \cos \theta)^2 = \sin^2(2\theta),$$

so the map is a doubling $\theta \mapsto 2\theta$ in the θ coordinate. The multiplier at the origin is 4 rather than 2, which suggests that the linearisation involves θ^2 ; and indeed

$$\psi_4(x) = (\arcsin \sqrt{x})^2. \quad (2.46)$$

Using $\arcsin(2u\sqrt{1 - u^2}) = 2 \arcsin u$ with $u = \sqrt{x}$, on an appropriate real interval, gives $\psi_4(f_4(x)) = (2 \arcsin \sqrt{x})^2 = 4\psi_4(x)$; and as $x \rightarrow 0$, $\arcsin \sqrt{x} \approx \sqrt{x}$, so $\psi_4(x) \approx x$ and $\psi'_4(0) = 1$.

2.A.3 Affine conjugacy to $z^2 + c$

Lemma 2.18. *The logistic map $f_r(x) = rx(1 - x)$ is affinely conjugate to $P_c(z) = z^2 + c$ with*

$$c = c(r) := \frac{2r - r^2}{4}. \quad (2.47)$$

Proof sketch. Seek $h(x) = ax + b$ with $h \circ f_r \circ h^{-1} = P_c$. The substitution $z = r(\frac{1}{2} - x)$ — equivalently $x = -z/r + \frac{1}{2}$, for $r \neq 0$ — yields a monic quadratic; if $x' = f_r(x)$ and $z' = r(\frac{1}{2} - x')$, direct substitution gives $z' = z^2 + c$ with c as in [\(2.47\)](#). $\square$

Under this conjugacy, $r = 2$ corresponds to $c = 0$ (the monomial z^2) and $r = 4$ to $c = -2$ (Chebyshev type; the value $c = -2$ is also reached at $r = -2$). Neither exceptional value lies in the image of $r \in (0, 1)$, which is $c \in (0, \frac{1}{4})$. As explained in [remark 2.17](#), this parameter count is a consistency check rather than the proof: the decisive point is that both exceptional cases sit over *repelling* fixed points, which the attracting window excludes structurally.

2.A.4 The worked examples inside the Becker–Bergweiler classification

[Theorem 2.14](#) says that a differentially algebraic Schröder function exists only over a repelling fixed point, and is then a Möbius transformation of $\exp(\alpha t^\ell)$, of $\cos(\alpha t^\ell + \beta)$, or of Weierstrass- $\wp$ material, with the underlying map Möbius-conjugate to $z^{\pm d}$, to $\pm T_d$, or to a Lattès map. The two derivations above realise the first two families exactly.

Case $r = 2$. Inverting (2.45), the relation $t = -\frac{1}{2} \ln(1 - 2x)$ gives

$$\varphi_2(t) = \psi_2^{-1}(t) = \frac{1 - e^{-2t}}{2},$$

an affine — hence Möbius — image of $\exp(-2t)$: the family $\exp(\alpha t^\varrho)$ with $\varrho = 1$, $\alpha = -2$. Correspondingly the substitution $u = 1 - 2x$ conjugates f_2 to the monomial $u \mapsto u^2$, and the linearised fixed point has multiplier 2, which is repelling, as theorem 2.14(i) requires.

Case $r = 4$. Inverting (2.46), the relation $t = (\arcsin \sqrt{x})^2$ gives

$$\varphi_4(t) = \psi_4^{-1}(t) = \sin^2 \sqrt{t} = \frac{1 - \cos(2\sqrt{t})}{2},$$

a Möbius image of $\cos(\alpha t^\varrho + \beta)$ with $\varrho = \frac{1}{2}$, $\alpha = 2$, $\beta = 0$; the fractional exponent is admissible precisely because $\cos(2\sqrt{t})$ is an entire function of t . Correspondingly $u = 1 - 2x$ conjugates f_4 to $T_2(u) = 2u^2 - 1$, the second Chebyshev polynomial, and the multiplier at the linearised fixed point is 4, again repelling.

Both examples therefore sit exactly where theorem 2.14 says all differentially algebraic Schröder functions must sit: over repelling fixed points, in the exponential and cosine families. For $r \in (0, 1)$ the fixed point is attracting, no member of the classification is available, and theorem 2.16 gives hypertranscendence of ψ_r — which is what underpins the main claims of this chapter about $A(p) = 2\Psi_{2p}(\frac{1}{2})$.

2.B Catalogue of exploratory formulae

This appendix records the remaining output of the closed-form searches described in section 2.7. The expressions below are approximations only. They have no inferential status whatever — they are records of a curve-fitting exercise, not proposed identities — and they are retained for one reason: they show how little a visually accurate fit says about the singular quantity $1/A(p)$.

The constrained rational candidate

Polynomial and rational searches can be constrained to the known endpoints $A(0) = \frac{1}{2}$ and $A(\frac{1}{2}) = 0$. One of the more economical rational candidates was $\hat{A}_1$ of (2.31), reproduced here for comparison:

$$\hat{A}_1(p) = \frac{\frac{1}{2}(1 - 2p)(1 + 2p)}{1 + \frac{1}{2}p - \frac{5}{2}p^2 - \frac{6}{5}p^3 + \frac{6}{5}p^4}, \quad \hat{A}'_1(\tfrac{1}{2}) = -\frac{2}{0.55} \approx -3.636.$$

It passes through both endpoints and misses the critical slope -4 required by remark 2.6. Matching the values while missing the slope is enough to make the relative error in A , and hence the error in $1/A$, unacceptable near criticality.

Unconstrained symbolic tree search

A less restricted symbolic tree search produced many longer formulae of similar visual quality. Two representative outputs were

$$\begin{aligned} \hat{A}_2(p) = & 0.31696448624690005 \log \left(\left| \cos \left(\left(\frac{1}{3.19801133824532} \right)^{1/4} \right. \right. \right. \\ & \left. \left. \left. + p \left[\sin \left(\cos(|p|(0.621865696665606 - p) \cos(e^{e^p})) \right) + |p| \right] \right) \right| \right) + 0.5982282661889831 \end{aligned} \quad (2.48)$$

and $\hat{A}_3$ of (2.32). Both reproduce the target curve to within the thickness of a plotted line across the whole interval (figure 2.3), and neither is interpretable in any useful sense. The characteristic signature of the search is on display: deep nestings of trigonometric and iterated exponential terms carrying a dozen or more fitted decimal constants, with no structural relationship to the recursion that defines A .

Spliced surrogates

Forcing the correct critical branch by splicing gives $\hat{A}_4$ of (2.33), which is a workable practical surrogate and nothing more. If a splice is wanted, the certified partial product (2.15) is preferable on the first branch and the proved asymptotic (2.21) on the second; that is the scheme adopted in section 2.12, and unlike $\hat{A}_4$ it comes with error control at every parameter.

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